Synthetic Stock Data via a Mixture of Experts Neural SDE: Modeling Fibonacci Levels

Alireza M. Javid, Sina Molavipour

Abstract

Generating synthetic financial time series that faithfully reproduce real-world market dynamics remains a fundamental challenge in computational finance. Although recent continuous-time generative models, particularly Neural Stochastic Differential Equations (Neural SDEs), successfully capture important statistical properties of financial data, they provide no explicit mechanism for incorporating practitioner-defined technical price geometry. We propose a Fibonacci-aware Neural SDE that augments the latent state with path-dependent distances to dynamically computed Fibonacci retracement levels. The drift and diffusion functions are parameterized using a Mixture-of-Experts (MoE) architecture whose specialized experts model three major market responses around technical price levels: elastic mean-reversion (*Bounce*), directional continuation (*Break*), and price consolidation (*Hover*). A gating mechanism and entropy regularization encourage interpretable regime specialization while preserving realistic price interactions around technical levels. We evaluate the proposed framework by comparing the generated trajectories with both classical stochastic baselines and real financial time series, and by assessing the utility of the generated data in a downstream classification task. Experimental results demonstrate improved statistical fidelity and show that classifiers trained solely on the generated data achieve performance approaching those trained on real data, highlighting the effectiveness of incorporating localized price geometry into continuous-time generative modeling. Our open-sourced code can be found at https://github.com/s1369m/synthetic_stock.

Keywords

Synthetic data generation, Neural SDE, Mixture of experts, Fibonacci retracement levels

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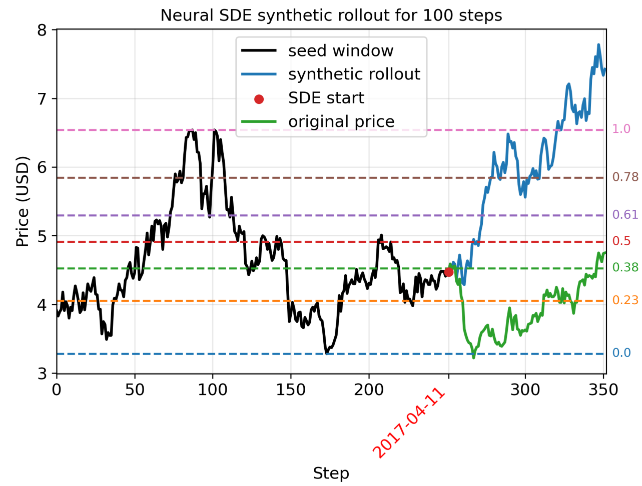

Figure 1: An example of synthetic stock path for 100 days generated by MoE Neural SDE with a 252 day lookback window, presenting one possible realistic price interactions with Fibonacci levels.

1 Introduction

Modeling financial time series has posed a fundamental challenge over the years for the machine learning community: standard quantitative models capture global statistical properties (like fat tails and volatility clustering), while real-world markets also exhibit localized dynamics arising from regime-switching human and algorithmic behaviors around key price levels. Therefore, market dynamics exhibit both continuous stochastic evolution and discrete, regime-dependent shifts driven by market participants. While continuous-time generative frameworks—most notably Neural Stochastic Differential Equations (Neural SDEs)—excel at modeling continuous stochastic dynamics, they typically provide no explicit inductive bias toward representing localized technical price-level structures. Conversely, market practitioners frequently monitor various support/resistance levels such as Fibonacci retracements, arguing that widespread attention to these levels may influence trading behavior. In this work, we bridge this gap by introducing a path-dependent Neural SDE constrained by a Mixture-of-Experts (MoE) gating architecture, explicitly embedding localized price geometry into continuous-time drift and diffusion dynamics. We use Fibonacci retracement levels as one representative example of practitioner-defined price geometry that can be incorporated into continuous-time stochastic generative models. Figure 1 illustrates an example of the generated synthetic stock path exhibiting realistic price interactions with the resulting Fibonacci levels.

Existing literature has largely explored these perspectives separately. Within quantitative finance, continuous-time models have traditionally focused on established "stylized facts" (such as heavy tails, volatility clustering, and stochastic volatility) [1, 4] while rarely incorporating practitioner-defined technical levels into generative stochastic models. In contrast, machine learning approaches that incorporate technical levels typically treat them as engineered input features for downstream classifiers or forecasting models (e.g., LSTMs or Random Forests) [5, 8, 16]. Meanwhile, generative approaches such as QuantGAN [20] and TimeGAN [21] learn temporal and statistical structure directly from historical data rather than explicitly encoding practitioner-defined price geometry. Consequently, existing generative architectures generally do not provide an explicit mechanism for conditioning continuous dynamics on localized price geometry.

Among continuous-time generative models, Neural SDEs have emerged as a particularly promising framework for financial time series due to their ability to model stochastic latent dynamics as well as irregularly sampled observations. Foundational work in Neural ODEs [3] established vector-field parameterizations of latent state evolution, which were subsequently extended to Neural CDEs [11] to handle continuous data updates via control path increments $dX(t)$. To capture intrinsic market noise and path-dependent uncertainty, Neural SDEs [13, 18] explicitly parameterize drift $f(\cdot;\theta_f)$ and diffusion $g(\cdot;\theta_g)$ terms driven by Wiener processes $W(t)$. Neural SDEs are commonly trained using numerical SDE solvers such as Euler–Maruyama [7], while theoretical guarantees for strong existence and uniqueness typically rely on Lipschitz-type conditions on the drift and diffusion coefficients [14]. Recent advances have focused on improving training stability; for instance, non-adversarial frameworks leverage signature kernel scores to prevent mode collapse during path generation [10]. In parallel, classical quantitative finance provides well-established stochastic priors, including Geometric Brownian Motion (GBM) [2] and Ornstein–Uhlenbeck (OU) dynamics [9, 19]. A small but growing body of work has begun integrating such priors into neural diffusion and continuous-time generative models as inductive biases for the forward or latent dynamics [12]. While these approaches improve consistency with established macro-level financial behavior, they remain focused on global statistical structure rather than localized price-level interactions. These limitations suggest that richer mechanisms are needed to model localized, regime-dependent behavior within continuous-time stochastic dynamics.

Mixture-of-Experts (MoE) architectures provide a natural framework for modeling heterogeneous and regime-dependent time series through input-dependent specialization. Early work demonstrated that MoEs can effectively combine local autoregressive models across distinct market regimes [22], while modern architectures employ learned gating networks to dynamically route inputs among specialized experts [6, 17]. The state-dependent routing induced by the gating mechanism also makes MoEs well suited for modeling locally varying dynamics within continuous-time systems. Building on this perspective, we introduce three specialized heads—*Bounce*, *Break*, and *Hover*—to form an economically interpretable gating mechanism representing different price interactions at levels.

To define these localized price interactions, we use Fibonacci retracement levels to construct a path-dependent feature space that directly conditions the drift and diffusion of a Neural SDE. Unlike conventional approaches that treat such levels as engineered features for downstream prediction, our formulation incorporates them into the continuous-time dynamics themselves. To the best of our knowledge, this is the first work to integrate Fibonacci-derived path-dependent geometry directly into a Neural SDE, providing an explicit bias toward localized price interactions while retaining the flexibility of continuous-time stochastic modeling.

We evaluate the proposed framework by comparing the generated trajectories against both classical stochastic baselines and real financial time series in terms of statistical fidelity. We further assess the practical utility of the generated data in a downstream classification task, demonstrating that models trained exclusively on synthetic data achieve performance statistically close to that of models trained on real data.

2 Fibonacci-aware Neural SDE

Generating synthetic financial time series that faithfully reproduce the statistical properties of real markets is a well-studied problem. However, existing generative models typically focus on standard stylized facts—such as volatility clustering and heavy tails—while neglecting the scale-invariant geometric patterns utilized by market practitioners. Specifically, price dynamics frequently exhibit localized mean reversion and volatility compression around Fibonacci retracement levels (e.g., 38.2%, 50.0%, and 61.8%) derived from recent swing highs and lows. Because these levels depend entirely on the historical trajectory of the asset, capturing their influence requires departing from strictly Markovian frameworks. We propose a novel, path-dependent Neural Stochastic Differential Equation (Neural SDE) architecture designed to synthesize stock data while explicitly preserving these structural Fibonacci properties.

To operationalize this, we augment the state space of a standard Neural SDE. Let X_t denote the asset price at time t, and let $X_{[t-w,t]} = \{X_s : s \in [t-w,t]\}$ represent the path history over a lookback window of length w. Based on $X_{[t-w,t]}$, we can deterministically compute a vector of rolling-window Fibonacci levels, $F_t^{(r)}$, where $r \in \mathcal{R}$ belongs to the popular set of ratios in quantitative finance:

$$\mathcal{R} = \{0, 0.236, 0.382, 0.500, 0.618, 0.786, 1.0\},$$

which updates dynamically as t grows and new local extrema are established. Specifically, Fibonacci levels are calculated as:

$$F_t = \begin{bmatrix} L_t \\ L_t + 0.236\delta_t \\ L_t + 0.382\delta_t \\ L_t + 0.500\delta_t \\ L_t + 0.618\delta_t \\ L_t + 0.786\delta_t \\ H_t \end{bmatrix}, \quad H_t = \max_{s\in[t-w,t]} X_s, \quad L_t = \min_{s\in[t-w,t]} X_s,$$

where H_t and L_t are simply the maximum and minimum of the price path within the window, and $\delta_t = H_t - L_t$ represents the total price range (the "swing") within the lookback window.

Let's define $D_t = (X_t - F_t)/\delta_t$ as the normalized distance of price at time t to each Fibonacci level. The continuous-time evolution of the synthetic asset is then governed by the following path-dependent SDE:

$$dX_t = \mu_\theta(X_{[t-w,t]}, D_t)dt + \sigma_\phi(X_{[t-w,t]}, D_t)dW_t, \tag{1}$$

where μ_θ and σ_θ are highly expressive neural networks parameterizing the drift and diffusion, respectively, and W_t is a standard

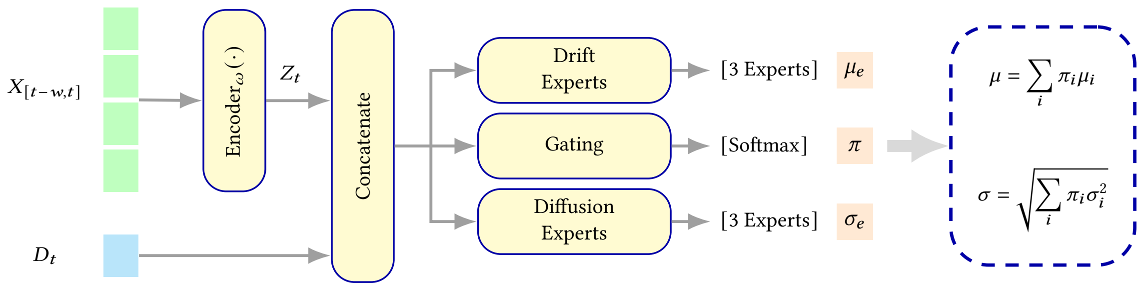

Figure 2: Architecture overview of the Path-Dependent MoE Neural SDE. Historical trajectory $X_{[t-w,t]}$ is encoded into a latent vector Z_t and concatenated ($\|$) with the distance vector D_t to form $[Z_t \parallel D_t]$, which routes simultaneously to the gating network and structural experts.

Brownian motion. By conditioning the neural networks on both the price history $X_{[t-w,t]}$ and the distance to the active levels in F_t, the model can learn the conditional distributions of price reversals, breakouts, and consolidations historically observed near these key zones. A high-level diagram of the proposed MoE architecture is shown in Figure 2.

2.1 Path-dependent MoE

To capture path dependency, we encode the continuous history $X_{[t-w,t]} = \{X_s : s \in [t-w,t]\}$ using a sequence encoder (e.g., GRU, LSTM, or Path Signature transform) into a compact latent embedding Z_t:

$$Z_t = \text{Encoder}_\omega(X_{[t-w,t]}) \tag{2}$$

A single drift and diffusion parameterization may average over heterogeneous local responses, potentially smoothing distinct behavioral regimes. To preserve distinct path regimes, specifically Bounce (reversal), Break (continuation), and Hover (consolidation), we parameterize the drift and diffusion using a Mixture of Experts (MoE) architecture governed by a gating network $\boldsymbol{\pi}_\psi(Z_t, D_t)$:

$$\boldsymbol{\pi}_\psi(Z_t, D_t) = \text{Softmax}\left(f_{\text{gate}}(Z_t, D_t)\right) \in \mathbb{R}^3, \tag{3}$$

where $\boldsymbol{\pi} = [\pi_{\text{bounce}}, \pi_{\text{break}}, \pi_{\text{hover}}]^T$ represents the probability of each regime.

To enforce domain-specific inductive biases and prevent mode collapse, we apply structural constraints to each expert model:

- Bounce Expert (μ_{bounce}): Models elastic mean-reversion away from the nearest Fibonacci level $r^* = \arg\min_r |D_t^{(r)}|$. Its drift direction is mathematically constrained to push the price away from $F_t^{(r^*)}$:

$$\mu_{\text{bounce}}(Z_t, D_t) = \text{sign}(D_t^{(r^*)}) \cdot \text{Softplus}\left(g_{\text{bounce}}(Z_t, D_t)\right) \tag{4}$$

- Break Expert (μ_{break}): Models directional momentum continuation through the level. Its drift direction is locked to the sign of recent price velocity $\Delta X_t = X_t - X_{t-\Delta t}$:

$$\mu_{\text{break}}(Z_t, D_t) = \text{sign}(\Delta X_t) \cdot \text{Softplus}\left(g_{\text{break}}(Z_t, D_t)\right) \tag{5}$$

- Hover Expert (μ_{hover}): Models price pinning and market indecision near support/resistance. Its drift magnitude is strictly bounded near zero:

$$\mu_{\text{hover}}(Z_t, D_t) = \gamma \cdot \tanh(g_{\text{hover}}(Z_t, D_t)), \quad \gamma \ll 1 \tag{6}$$

Then, the global drift μ_θ can be computed as the convex combination of the expert outputs:

$$\mu_\theta(Z_t, D_t) = \sum_{m \in \mathcal{M}} \pi_m(Z_t, D_t) \cdot \mu_m(Z_t, D_t), \tag{7}$$

where $\mathcal{M} = \{\text{bounce, break, hover}\}$.

Note that constraining μ_{bounce} with $\text{sign}(D_t^{(r^*)})$ and μ_{break} with $\text{sign}(\Delta X_t)$ imposes the intended directional constraints on the expert drifts, reducing the burden on the neural network to "rediscover" momentum and mean-reversion from scratch.

Similarly, the gating network defines an RMS aggregation of expert diffusion coefficients, yielding a single effective SDE rather than an explicit mixture distribution:

$$\sigma_\phi(Z_t, D_t) = \sqrt{\sum_{m \in \mathcal{M}} \pi_m(Z_t, D_t) \cdot \sigma_m^2(Z_t, D_t)}, \tag{8}$$

where σ_{hover} is constrained to small values ($\sigma_{\text{hover}} \approx \sigma_{\min}$) to represent reduced price variability during consolidation. Here, we set:

- $\sigma_{\text{bounce}} = \text{Softplus}(h_{\text{bounce}}(Z_t, D_t))$,
- $\sigma_{\text{break}} = \text{Softplus}(h_{\text{break}}(Z_t, D_t))$,
- $\sigma_{\text{hover}} = \sigma_{\min} + \gamma \cdot \text{Sigmoid}(h_{\text{hover}}(Z_t, D_t))$.

Note that each of the functions f, g, and h are parametrized by some form of neural networks, modeling gating, drift, and diffusion, respectively. The choice of architecture for these networks can vary depending on the complexity of the data and the task among MLP, transformer, LSTM, etc.

2.2 Parameter Estimation and Entropy Regularization

The local transition over an infinitesimal step Δt remains Gaussian relative to the augmented state (Z_t, D_t). The step-wise negative log-likelihood for X_{i+1} given historical path window $X_{[t_i-w,t_i]}$ is:

$$\begin{aligned}\mathcal{L}_i(\theta, \phi, \psi) =& \frac{1}{2}\log\left(2\pi\sigma_\phi^2(Z_i, D_i)\Delta t\right) \\ &+ \frac{(X_{i+1} - X_i - \mu_\theta(Z_i, D_i)\Delta t)^2}{2\sigma_\phi^2(Z_i, D_i)\Delta t}\end{aligned} \tag{9}$$

To prevent the gating network from collapsing prematurely into a single dominant expert during early training phases, we augment the negative log-likelihood objective with a categorical entropy

Algorithm 1 Path-Dependent Mixture-of-Experts Neural SDE Training

Input: Historical dataset of path-target pairs $\mathcal{D} = \left\{ \left(X^{(m)}_{[t-w,t]}, X^{(m)}_{t+1} \right) \right\}_{m=1}^{M}$
Lookback window size w, time step $\Delta t = 1$, batch size N, and training epochs E
Fibonacci ratio set $\mathcal{R} = \{0, 0.236, 0.382, 0.500, 0.618, 0.786, 1.0\}$
Market regime set $\mathcal{M} = \{\text{bounce}, \text{break}, \text{hover}\}$
Entropy regularization scaling hyperparameter β, hover bound scale γ, and minimum volatility $\sigma_{\min}$

Initialization: Set trainable parameters ω for Encoder$_\omega$; ψ for f_{gate}; θ for expert drift networks $\{g_{\text{bounce}}, g_{\text{break}}, g_{\text{hover}}\}$; and ϕ for expert diffusion networks $\{h_{\text{bounce}}, h_{\text{break}}, h_{\text{hover}}\}$. Clear gradients.

for epoch $= 1 \ldots E$ **do**
  for each mini-batch $\mathcal{B} = \left\{ \left(X^{(i)}_{[t-w,t]}, X^{(i)}_{t+1} \right) \right\}_{i=1}^{N} \subset \mathcal{D}$ **do**
    $\mathcal{L}_{\text{NLL}} \leftarrow 0$, $\mathcal{H}_{\text{total}} \leftarrow 0$
    # MoE Loss and Entropy Accumulation over Batch
    for each sample $i \in \{1, \ldots, N\}$ in mini-batch $\mathcal{B}$ **do**
      Extract history $X^{(i)}_{[t-w,t]}$, current price $X^{(i)}_t$, and target price $X^{(i)}_{t+1}$
      # Compute Fibonacci features for sample i
      $H_t \leftarrow \max(X^{(i)}_{[t-w,t]})$, $L_t \leftarrow \min(X^{(i)}_{[t-w,t]})$, $\delta_t \leftarrow H_t - L_t$
      $F_t \leftarrow [L_t + r \cdot \delta_t]$, where $r \in \mathcal{R}$
      $D_t \leftarrow (X^{(i)}_t - F_t)/\delta_t$
      Encode path history: $Z_t \leftarrow \text{Encoder}_\omega \left(X^{(i)}_{[t-w,t]} \right)$
      # Evaluate Transition under MoE
      Compute gating distribution: $\boldsymbol{\pi}_t \leftarrow \text{Softmax}(f_{\text{gate}}(Z_t, D_t))$
      Find nearest Fibonacci level index: $r^* \leftarrow \arg\min_r |D^{(r)}_t|$
      Compute velocity direction: $\Delta X_t \leftarrow X^{(i)}_t - X^{(i)}_{t-1}$
      Compute drift experts $\mu_{\text{bounce}}, \mu_{\text{break}}, \mu_{\text{hover}}$ and diffusions $\sigma_{\text{bounce}}, \sigma_{\text{break}}, \sigma_{\text{hover}}$ as per Section 2.1.
      Aggregate parameters: $\mu_\theta(Z_t, D_t) \leftarrow \sum_m \pi_{t,m}\mu_m$ and $\sigma_\phi(Z_t, D_t) \leftarrow \sqrt{\sum_m \pi_{t,m}\sigma_m^2}$
      # Accumulate step likelihood and entropy
      $\mathcal{L}_{\text{NLL}} \leftarrow \mathcal{L}_{\text{NLL}} + \frac{1}{2}\log(2\pi\sigma^2_\phi(Z_t, D_t)) + \frac{\left(X^{(i)}_{t+1} - X^{(i)}_t - \mu_\theta(Z_t, D_t) \right)^2}{2\sigma^2_\phi(Z_t, D_t)}$
      $\mathcal{H}_{\text{total}} \leftarrow \mathcal{H}_{\text{total}} - \sum_{m \in \mathcal{M}} \pi_{t,m} \log(\pi_{t,m} + \epsilon)$
    end for
    # Regularized Objective Optimization
    $\mathcal{L}_{\text{total}} \leftarrow \frac{1}{N}\mathcal{L}_{\text{NLL}} - \beta\frac{1}{N}\mathcal{H}_{\text{total}}$ (Batch-normalized loss with entropy regularization)
    Update parameters via backpropagation: $(\omega, \psi, \theta, \phi) \leftarrow \text{Adam}(\nabla_{\omega,\psi,\theta,\phi}\mathcal{L}_{\text{total}})$
  end for
end for

Output: Optimized encoder Encoder$_\omega$, gating network f_{gate}, drift experts μ_θ, and diffusion experts σ_ϕ.

regularizer on the gating distribution $\boldsymbol{\pi}$:

$$\mathcal{H}(\boldsymbol{\pi}_i) = -\sum_{m \in \mathcal{M}} \pi_{i,m} \log(\pi_{i,m} + \epsilon). \tag{10}$$

Then, the total objective function minimized over the trajectory is given by:

$$\mathcal{L}_{\text{total}} = \frac{1}{N}\sum_{i=0}^{N-1} \mathcal{L}_i(\theta, \phi, \psi) - \beta \cdot \frac{1}{N}\sum_{i=0}^{N-1} \mathcal{H}(\boldsymbol{\pi}_i), \tag{11}$$

where $\beta > 0$ controls the strength of entropy regularization, encouraging uniform expert exploration early in optimization before annealing to allow sharp regime selection. The training algorithm for the proposed MoE architecture is detailed in Algorithm 1.

3 Experimental Results

In this section, we carry out several experiments to evaluate the quality of the synthetic data generated by MoE architecture in various settings. First, we compare statistical fidelity of our synthetic data and compare that against other benchmarks. Then, we define a downstream task as linear trend classification for stock prices and evaluate weather models trained on synthetic data are able to achieve a similar performance as when trained on real data.

3.1 Experimental Setup

The MoE architecture in all experiments of this section uses a temporal encoder consisting of a multi-layer perceptron with hidden layer dimensions of 64 and 32 units, using ReLU activations. The gating network employs a SwiGLU-based feedforward architecture with 64 hidden units, LayerNorm layers, and residual connections, outputting selection probabilities across the 3 regimes. Each expert head is implemented as a two-layer feedforward network with a hidden dimension of 32 units, LayerNorm, and SiLU activation. The model was trained using a batch size of 256 for a maximum of 20 epochs. Furthermore, the entropy regularization weight β is dynamically adjusted using a warmup and cosine decay scheduler.

To isolate the benefit of mixture-of-experts gating, we construct a monolithic MLP baseline sharing the same backbone dimensions (64 and 32 hidden units, ReLU activations). The MLP directly parameterizes μ via a linear head and σ via a Softplus head without any gating mechanism. Additionally, we implement a PatchTST-style Transformer baseline that segments the 252-day lookback window into 42 non-overlapping 6-day patches. Embedded tokens are processed through a single-layer, 2-head Transformer encoder ($d_{\text{model}} = 32$), mean-pooled, and concatenated with current-step features to predict drift μ and diffusion σ.

The dataset used for training the Neural SDE contains rolling lookback windows of length 252 days constructed from 50 random US stocks prices between 2014-12-31 to 2017-12-31, with a random 80-20% split for train and validation sets, respectively. The dataset used for testing the downstream classifier in Section 3.3 contains rolling lookback windows of length 21 days and lookforward window of length 5 days constructed from 500 random US stocks prices between 2017-12-31 to 2020-12-31 to avoid data leakage across time during synthetic generation and downstream evaluation. All 252-day lookback window samples are min-max normalized before feeding into Neural SDE and the classifier.

3.2 Statistical Evaluation

To evaluate whether the Neural SDE captures path-level financial dynamics beyond one-step predictive accuracy, we generate rollout trajectories against observed market data. Specifically, we sample 10,000 lookback windows of length 252 days across 50 US stocks spanning 2014-12-31 to 2017-12-31. For each window, we generate 20 Monte Carlo (MC) rollout paths over a 100-day forecast horizon (an example rollout is illustrated in Figure 1).

As a baseline, we generate matching MC rollouts by adapting Vasicek/OU dynamics as a generic mean reverting benchamrk for price trajectories, with fixed parameters ($\kappa = 0.015, \mu = 0.50, \sigma = 0.015$) being calibrated via maximum likelihood estimation (MLE) on historical price paths. We compare models across three key metrics: mean daily return ($\bar{r}$), daily return volatility (σ_r), and maximum drawdown (MDD). While $\bar{r}$ and σ_r measure marginal return properties, MDD evaluates explicit path-dependent downside risk.

To prevent stocks with larger numbers of windows from biasing the results, evaluation is aggregated at the stock level. For each stock i, metrics are averaged across MC realizations and forecast origins, yielding a paired cross-sectional tuple $(s_i^{\text{orig}}, s_i^{\text{MoE}}, s_i^{\text{Vas}})$ across the $N = 50$ stocks for each metric s. Then, we average these metrics over our total sample size of 50 stocks to calculate the overall cross-sectional mean for each case of original, MoE, and Vasicek.

Figure 3 illustrates the resulting empirical densities for MoE architecture. Across all metrics, the Neural SDE cross-sectional mean aligns substantially closer to the original ground truth than the Vasicek process. We also repeat the same experiment using a standard MLP model instead of MoE, shown in Figure 4. We can observe that MoE achieves a reduced distance to the empirical mean by 0.004% vs. 0.126% for $\bar{r}$, 0.241% vs. 1.355% for σ_r, and 4.097% vs. 9.409% for MDD (evaluated against a standard MLP model baseline). The accurate recovery of MDD confirms that the Neural SDE faithfully models temporal accumulation of losses.

Table 1: Comparison of Jensen–Shannon distance between histograms of Fibonacci interactions.

Regime	MoE Synthetic	MLP Synthetic	Vasicek
Bounce	**0.04**	0.11	0.30
Break	**0.13**	0.19	0.42
Hover	**0.11**	0.18	0.42

3.2.1 Interaction with Fibonacci levels. To check whether the synthetically generated paths from Neural SDE hold properties similar to original price movements, we examine the statistics of the path when interacting with Fibonacci levels as a deterministic finite set of states.

Let's assume an interaction is triggered when a normalized price trajectory enters a ±0.02 normalized band around a Fibonacci level, transitioning from *Idle* to *Approaching*. If the trajectory immediately reverses, the event classified as an immediate *Bounce*. If it remains within the band for $N_{\text{dwell}} \geq 3$ days, it enters a *Consolidating* state. From there, subsequent movement yields one of three terminal outcomes: a continuation through the level (*Break*), a delayed reversal (*Bounce*), or an interaction duration exceeding $N_{\max} = 20$ days (*Timeout*).

Consequently, two interaction paths are possible:

$$\text{Idle} \rightarrow \text{Approaching} \rightarrow \text{Finished (Bounce)},$$

or

$$\text{Idle} \rightarrow \text{Approaching} \rightarrow \text{Consolidating (Hover)}$$
$$\rightarrow \text{Finished (Break, Bounce, or Timeout)}.$$

We evaluate this FSM across 50 stocks and 10 forecast origins, generating 20 Monte Carlo rollouts per setup. For each of the seven Fibonacci levels, we aggregate event frequencies into categorical outcome distributions. This histogram is then compared with the original price data and generated Vasicek paths as a benchmark, to evaluate how accurately the MoE synthetic data is capturing the interactions with Fibonacci level.

Figure 5 compares the resulting event distributions against observed market data for MoE and Vasicek models. To quantify distribution alignment, we compute the Jensen-Shannon (JS) distance between the empirical event distributions of the true and synthetic paths. We also repeat the same experiment using the standard MLP model and report the results in Table 1. We can observe that MoE consistently yields lower JS divergence than both the Vasicek process and the MLP baseline, confirming its ability to capture complex boundary-interaction dynamics in financial time series.

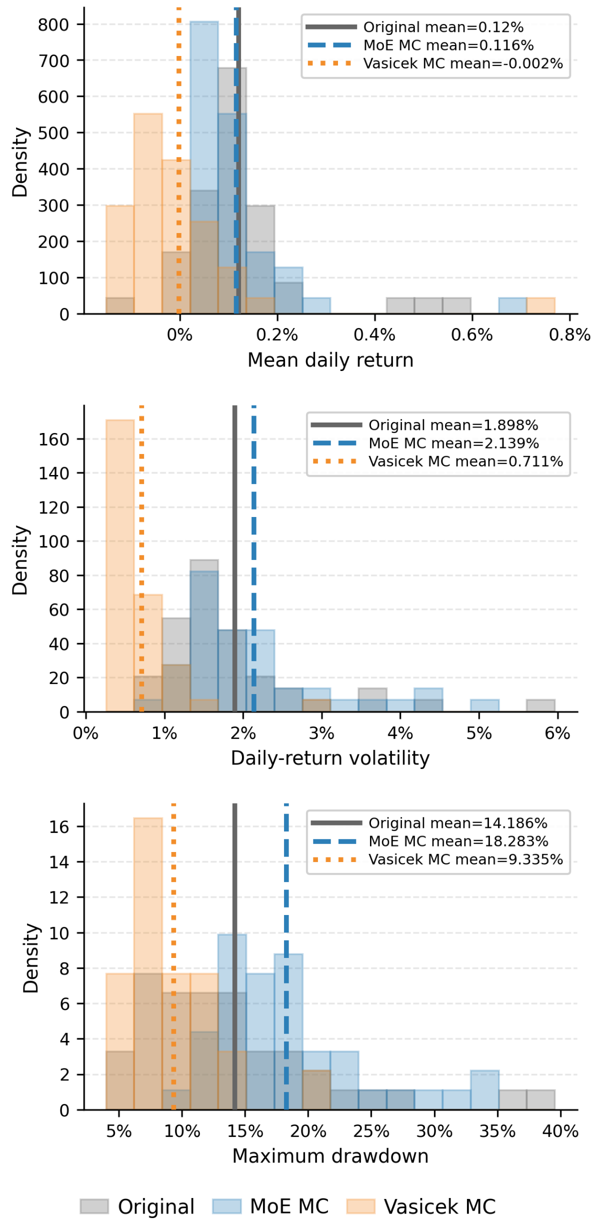

Figure 3: Cross-stock distributions of mean daily return, daily-return volatility, and maximum drawdown for the original data, MoE simulations, and Vasicek simulations. Each observation is aggregated across forecast dates and 20 Monte Carlo paths for each stock seed window.

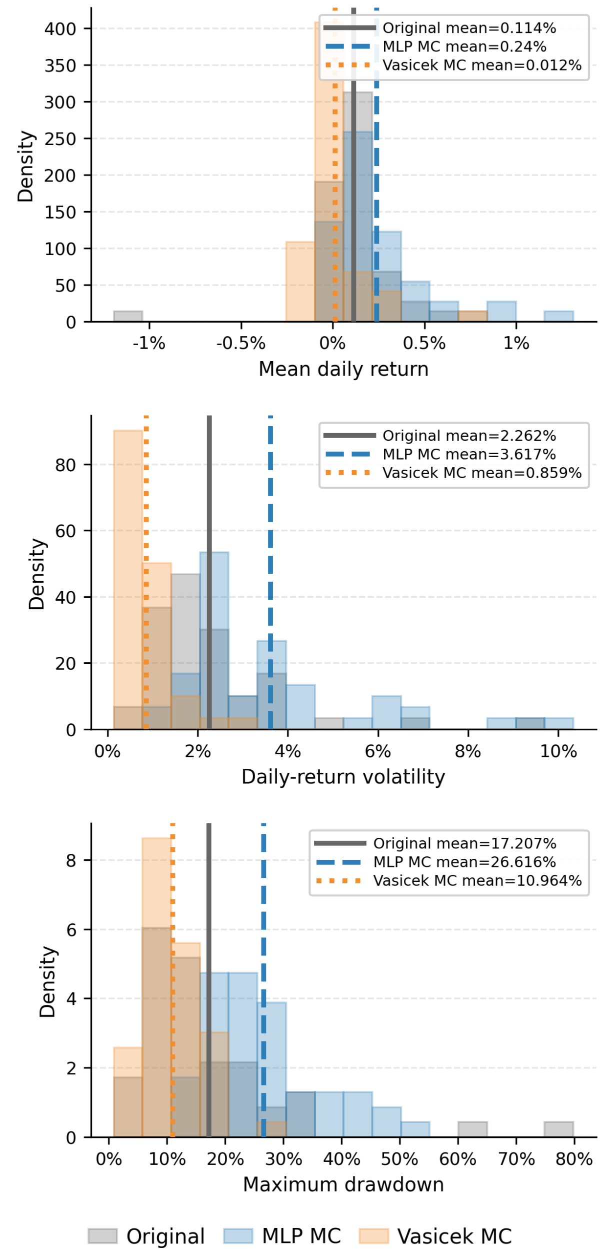

Figure 4: Cross-stock distributions of mean daily return, daily-return volatility, and maximum drawdown for the original data, MLP simulations, and Vasicek simulations. Each observation is aggregated across forecast dates and 20 Monte Carlo paths for each stock seed window.

3.3 Downstream Task Evaluation

To evaluate the impact of synthetic data share and architectural variations on financial forecasting performance, we deploy a 3-class path-dependent classification task. The objective is to predict price barrier crossings over a forward horizon of $N = 5$ trading days using a lookback window of $M = 21$ trading days. First, a linear trend model, $L(t) = \alpha + \beta t$, and its corresponding 95% confidence interval $\mathrm{CI}(t) = L(t) \pm 1.96\sigma$, are fitted on daily prices across the 21-day lookback window. Based on price movement in the subsequent 5-day horizon, targets are labeled as UpTouch (price first touches the upper CI boundary), DnTouch (price first touches the lower

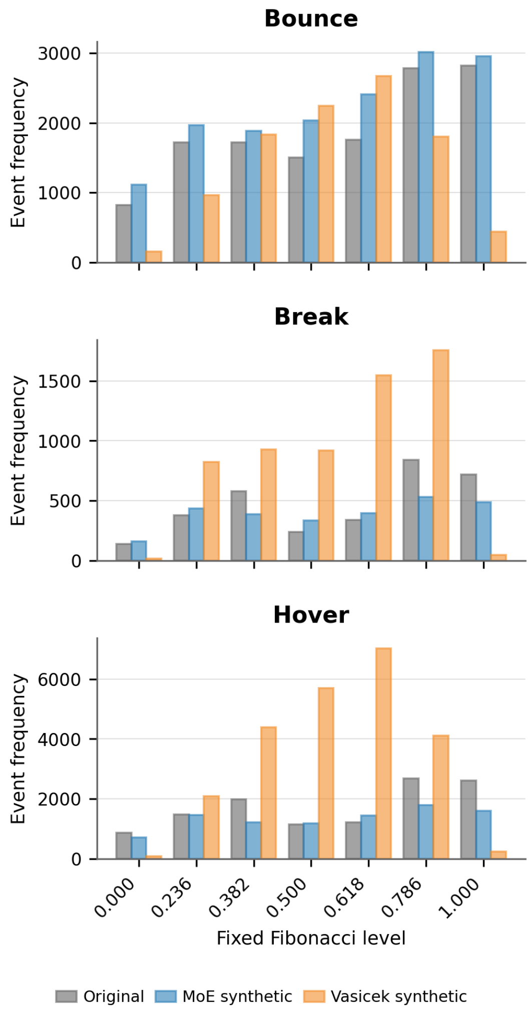

Figure 5: Comparison between MoE , Vasicek, and original stock price in interaction with Fibonacci levels.

CI boundary), or NoTouch (price remains strictly within the CI band). To ensure statistical stability each configuration is evaluated using an ensemble of 10 independent model instances. We examine performance across two key dimensions:

(1) Synthetic Data Share: systematically varying the proportion of synthetic training samples from 0% (real data only) to 100% (fully synthetic data) in increments of 25%.
(2) Model Architecture: comparing a standard Multi-Layer Perceptron (MLP) baseline against a Mixture of Experts (MoE) and a patch-based transformer (PatchTST) which is inspired by [15] as the name suggests.

Models are evaluated using train and testing accuracy as well as normalized confusion matrices across all three classes, with mean accuracy and standard deviation reported over the ensemble runs.

The empirical results in Table 2 and 3 demonstrate a distinct trade-off between the injection of synthetic data and class-wise predictive reliability. When scaling the synthetic data share from 0% to 100%, MoE maintains robust overall performance without severe degradation, confirming that the synthetic dataset successfully captures the underlying dynamics of the real price distributions. Interestingly, while the classification accuracy for the NoTouch class exhibits a minor decline as synthetic share increases (dropping from 48.1% to 43.2%), the model achieves its highest precision on directional barrier events (UpTouch and DnTouch) when trained purely on synthetic data (71.6 ± 2.5%, compared to 70.2 ± 2.3% for real data). This suggests that synthetic generation may act as an effective regularizer or oversampling mechanism for extreme boundary-touching market regimes or improving the performance of tail event prediction tasks.

Table 2: Model performance across varying synthetic data proportions (Total $N = 60$k).

Model (Synthetic%)	Train Acc (%)	Test Acc (%)
Real Data (0%)	**62.78 ± 0.14**	**61.88 ± 0.25**
MoE (25%)	62.23 ± 0.20	61.87 ± 0.33
MoE (50%)	62.12 ± 0.21	61.85 ± 0.29
MoE (75%)	61.97 ± 0.15	61.86 ± 0.32
MoE (100%)	**61.70 ± 0.09**	**61.83 ± 0.18**
MLP (100%)	60.34 ± 0.07	61.54 ± 0.24
PatchTST (100%)	61.62 ± 0.13	61.70 ± 0.28

Comparing across model architectures under the fully synthetic regime (100% synthetic data), MoE demonstrates superior overall fidelity and generalizability over standard neural baselines. As shown in Table 2, MoE achieves a test accuracy of 61.83 ± 0.18%, closely matching the real-data baseline (61.86±0.25%) while outperforming both PatchTST (61.70 ± 0.28%) and MLP (61.54 ± 0.24%). Notably, MLP exhibits significant training degradation (60.34 ± 0.07%), suggesting an inability to properly capture complex temporal dynamics without regime-specific structural priors. Examining the class-wise breakdowns in Table 3, MoE (100%) achieves the highest accuracy on the downward barrier touch class (*DnTouch*) at 70.6 ± 2.7%, compared to 69.8 ± 2.3% for MLP and 68.7 ± 3.6% for PatchTST. While PatchTST maintains slightly higher retention on the central *NoTouch* class (43.5 ± 3.8%), MoE offers a more balanced trade-off across directional extreme events without sacrificing one class. Furthermore, under the same experimental setup, the proposed MoE requires only 8.36 s/epoch for training, compared with 20.27 s/epoch for PatchTST, corresponding to approximately 2.42× faster training.

4 Conclusion

In this paper, we introduced a path-dependent Mixture-of-Experts (MoE) Neural Stochastic Differential Equation (SDE) framework designed to model financial asset dynamics near key technical levels. By augmenting the state space with dynamic Fibonacci distance features, our architecture explicitly models three distinct market regimes: *Bounce*, *Break*, and *Hover* while preserving statistical characteristics of the original stock data. We also showed that models trained on synthetic trajectories generated by our MoE Neural SDE maintain a remarkably stable test accuracy even at 100% synthetic

Table 3: Normalized Confusion Matrices (Mean ± Std in %) across synthetic data shares for each class.

Model (Synthetic%)	Predicted Class (%)		
	DnTouch	NoTouch	UpTouch
Real Data (0%)	**67.3 ± 3.9**	22.7 ± 3.7	10.0 ± 1.2
	25.8 ± 3.9	**48.1 ± 3.9**	26.1 ± 2.3
	8.8 ± 1.7	21.0 ± 2.1	**70.2 ± 2.3**
MoE (25%)	67.8 ± 3.1	22.1 ± 4.0	10.1 ± 1.9
	26.1 ± 3.2	47.4 ± 6.0	26.5 ± 4.4
	8.9 ± 1.2	20.7 ± 4.1	70.4 ± 4.0
MoE (50%)	70.9 ± 2.2	19.0 ± 2.7	10.1 ± 1.5
	29.4 ± 2.5	44.2 ± 4.7	26.5 ± 3.6
	10.0 ± 1.0	19.5 ± 3.4	70.5 ± 3.3
MoE (75%)	69.7 ± 3.3	19.4 ± 3.9	10.9 ± 1.3
	28.0 ± 3.3	43.4 ± 5.6	28.5 ± 3.2
	9.6 ± 1.3	18.0 ± 3.3	72.5 ± 2.7
MoE (100%)	**70.6 ± 2.7**	18.7 ± 2.6	10.7 ± 0.8
	29.0 ± 2.9	43.2 ± 3.3	27.8 ± 1.9
	9.8 ± 1.0	18.6 ± 1.8	71.6 ± 1.6
MLP (100%)	69.8 ± 2.3	19.0 ± 2.1	11.3 ± 1.1
	28.4 ± 2.6	41.9 ± 3.1	29.7 ± 2.6
	9.5 ± 1.1	17.3 ± 2.2	**73.2 ± 2.4**
PatchTST (100%)	68.7 ± 3.6	20.4 ± 2.9	10.9 ± 1.4
	27.3 ± 3.9	**43.5 ± 3.8**	29.2 ± 3.2
	9.4 ± 1.8	17.7 ± 2.3	72.9 ± 2.9

data share, narrowing the train–test generalization gap while improving class-wise recall for directional extreme events

Limitations and future work. A current limitation of the proposed framework is the stability of long-horizon rollouts. While the model produces stable trajectories over the 100-step generation horizon considered in our experiments, substantially longer rollouts can expose the learned drift and diffusion functions to rare states that are poorly represented in the training distribution. Since the SDE is recursively evaluated on its own generated states, such out-of-distribution excursions can accumulate and lead to increasingly extreme trajectories, including unrealistically large or negative prices. A promising direction for addressing this issue is to incorporate multiple temporal lookback windows into the state representation, allowing the drift and diffusion to condition on both short- and long-term historical context as in:

$$(X_{[t-252,t]}, X_{[t-756,t]}, X_{[t-2520,t]}, \tilde{D}_t) \to \{\mu_\theta, \sigma_\phi\} \to X_{t+1},$$

where $\tilde{D}_t$ contains the collective Fibonacci distances for all three windows. Future work will investigate such multi-scale path representations for more stable long-horizon generation, as well as extensions to multi-asset joint SDEs for capturing cross-asset dependencies and applications in portfolio risk management, including Value-at-Risk (VaR) estimation and xVA computations.

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